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Metabolism of apolipoproteins AI and AII in subjects carrying similar apoAI mutations, apoAI Milano and apoAI Paris

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Abstract

ApoAI Milano (AI_M) and apoAI Paris (AI_P) are mutant forms of apoAI in which cysteine is substituted for arginine at residues 173 and 151 respectively leading to the formation of homodimers and heterodimers with apoAII. Heterozygous subjects with these mutants are characterized by low levels of plasma HDL cholesterol and apoAI. The present study analyzed the metabolism of the different complexes of apoAI in three subjects, two AI_M and one AI_P, using a primed-constant infusion of trideuterated leucine. In AI_M carriers, the mutant form was almost equally distributed in AI_M dimer, AI_M:AII heterodimer and the monomer, whereas, in the AI_P subject, the mutant apoAI was essentially in the $apoAI_P$:AII complex. Normal apoAI was low in the AI_M subjects (20 and 16 mg/dl) but very low in the AI_P subject (0.3 mg/dl). In the AI_M subjects, the low levels of apoAI were due to a rapid catabolism with a normal synthetic rate. However, the apoAI kinetics were heterogeneous with a rapid catabolism of the AI_M:AII complex (FCR of 0.430 and 0.401 day^{-1}) and the AI_M monomer (FCR of 0.570 and 0.406 day^{-1}) whereas the AI_M dimer was catabolized slowly (FCR of 0.114 and 0.118 day^{-1}). In contrast, AI_P was catabolized relatively slowly with a FCR of 0.263, 0.182 and 0.258 day^{-1} for AI_P homodimer, $apoAI_P$:AII heterodimer and AI_P monomer. In the three subjects, normal apoAI was catabolized quickly, with an FCR of 0.805 and 0.601 day^{-1} in AI_M carriers and 0.526 day^{-1} in the AI_P carrier. Therefore, the low level of apoAI in the AI_P carrier is caused by a low production rate of apoAI, particularly of normal apoAI. In conclusion, apoAI is kinetically heterogeneous in AI_M and in AI_P subjects. Moreover, the two mutations lead to significant differences in the kinetic behavior of mutant apoAI depending on its inclusion in its complexes. © 2000 Elsevier Science Ireland Ltd. All rights reserved.

Keywords: Apolipoprotein AI; Hypoalphalipoproteinemia; Kinetics; Stable isotope

1. Introduction

Very low plasma levels of high density lipoproteins (HDL) can be due to a variety of different defects, including abnormal biosynthesis of apolipoprotein (apo) AI, the major protein constituent of plasma HDL [1], the production of structural variants of apoAI, or increased HDL catabolism such as in Tangier disease or in LCAT deficiency [2,3]. Epidemiological studies have shown that plasma concentrations of HDL are inversely correlated with the risk of premature atherosclerotic heart disease [4–6], but a few subjects with very low levels of HDL do not all seem to be predisposed to accelerated atherosclerosis [7]. Among clinical cases of hypoalphalipoproteinemia without susceptibility to an increased risk of CHD is that of subjects carrying a mutant form of apoAI, apoAI Milano ($apoAI_M$), which has been reported by Franceschini et al. [7] and in which a cysteine is substituted for an arginine at amino acid position 173 (R173C) [8].

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Recently, another mutation of apoAI, apoAI Paris ($apoAI_P$), was described in a French family in which the proband had a severe hypoalphalipoproteinemia. The mutant form of $apoAI_P$ had a cysteine substituted for an arginine at position 151 (R151C) and no sign suggesting a premature occurrence of atherosclerosis or CHD was found either in the proband, or in the family over three generations [9].

Because of the abnormal presence of a cysteine residue, both AI_M and AI_P form disulfide-linked dimers such as a heterodimer with apoAII or a homodimer. Therefore, HDL particles display complex features with monomers, homo- and heterodimers [9,10]. In addition, both mutants have one fewer net-positive charge than normal apoAI as a consequence of the loss of a positively charged arginine. This allows visualization of both normal and mutant apo-AI with isoelectric-focusing gel electrophoresis of plasma [8,9].

The kinetic etiology of the decreased plasma levels of apoAI in AI_M subjects was previously studied by radiolabeling normal and mutant $apoAI_M$. These studies showed that decreased apoAI levels are caused by the rapid catabolism of apoAI with a normal production rate of both the normal and mutant forms of apoAI [11]. $ApoAI_M$ also appeared to be catabolized much more rapidly as a monomer than as a dimer. However, in that study, the different complexes of mutant apoAI (heterodimer and homodimer) were not analyzed separately and the metabolism of apoAII was not studied further. In the present study, we have therefore investigated the metabolism of mutant apoAI present in different protein complexes, normal apoAI and apoAII in heterozygous AI_M and AI_P carriers using endogenous labeling of apoAI and apoAII with stable isotopes.

2. Materials and methods

2.1. Subjects

The two $apoAI_M$ subjects analyzed in the present study were identified during the screening of asymptomatic family members of the proband for $apoAI_M$ [12]. Subject 1 was a 52-year-old male and subject 2 was his 28-year-old son. These two subjects were heterozygous for mutation $apoAI_M$ (R173C). Subject 1 is identified as subject VI-171 and subject 2 as VII-171 in the pedigree published in Ref. [13]. Subject 3, a 43-year-old male, is the proband for $apoAI_P$ and was referred to one of us (GT) because of very low plasma HDL-cholesterol levels. ApoAI Paris was identified by sequence analysis of the apoAI gene which showed a heterozygous mutation in the apoAI gene which causes a cysteine for arginine replacement at residue 151 (R151C) [9]. None of the subjects smoked, nor did they drink alcohol excessively or take medication. The physical examination was normal except for subject 1 who was overweight (Body Mass Index: 36.7 kg/m^2) while the other two subjects' weights were normal (BMI: 24.4 and 24.5 kg/m^2). Control subjects were 10 non smoking normolipidemic volunteer males in good health. The study protocol was approved by the Ethic Committee (Comité Consultatif de Protection des Personnes dans la Recherche Biomédicale) of the University Hospital of Lille. The subjects gave informed written consent.

2.2. Kinetic study protocol

A stable isotope kinetic study using [5,5,5] $2H_3$-leucine (D3-leucine) was carried out in the Center for Clinical Investigation at the University Hospital in Lille, France. D3-leucine (Isotec, Miamisbourg, USA) with three deuterium atoms positioned in one of the methyl groups of the amino acid leucine was dissolved in a 0.9% NaCl solution, sterile filtered and tested for pyrogenicity (by the Limulus test) and sterility before use. After a 12-h fast, subjects were given a priming bolus injection of 8 μmol/kg of D3-leucine immediately followed by a constant infusion of 10 μmol/kg body wt per h of D3-leucine for 12 h. Blood samples were drawn into EDTA Vacutainer tubes from the opposite arm before the priming bolus, then every 15 min for the first hour, every 30 min for the second hour, then at hourly intervals until the end of the infusion. The subjects fasted during the infusion. Each time, plasma was immediately separated by centrifugation (2500 rev./min) for 20 min at 4°C. The plasma was kept at 4°C and sent to the laboratory (Department of Atherosclerosis, Institut Pasteur, Lille) where lipoprotein fractionation was started with 3 h of isolation.

VLDL (density < 1.006 g/ml) and HDL (1.063–1.210 g/ml) were separated by sequential ultracentrifugation [14] in a Beckman TL100 table centrifuge. VLDL and HDL were washed once at the upper density and were then exhaustively dialyzed against 5 mM NH_4HCO_3, pH 8.2, at 4°C. VLDL was isolated from all blood samples whereas HDL was only separated from plasma obtained at hourly intervals. The different apolipoproteins were purified by discontinuous SDS-PAGE (4–21%) according to the procedure of Laemmli [15]. For VLDL, gels were directly stained by R250 Coomassie Brilliant Blue and destained with a solution of methanol, acetic acid, water (30/58/12, v/v/v). Apo B100 was identified by comparing with known standards and was then cut out from the gels. Electrophoresis of HDL apolipoproteins in the presence of SDS and in non-reducing conditions first separated complexes of different molecular weights. This electrophoresis enabled us to distinguish three bands which were subsequently identified as apoAI dimer, apoAI/apoAII heterodimer and apoAI monomer (Fig. 1). Bands were detected by placing gels immediately after completion

of electrophoresis into 10–12 vol. of 4 M Na acetate [16]. Every protein band was excised with a razor blade and electroeluted for 4 h. The eluates were dialyzed and lyophilized. The eluted proteins corresponding to apoAI dimers and apoAI/apoAII complex were run on SDS-PAGE under reducing conditions and stained by Coomassie Blue and destained as above. So, apoAI from the apoAI dimer appeared as a monomer clearly separated from albumin (the apoAI dimer had indeed an apparent molecular weight similar to that of albumin and was contaminated by this protein) and the apoAI and the apoAII could be separated from the apoAI/apoAII heterodimer. The eluate corresponding to the apoAI monomer was solubilized for isofocalisation gel electrophoresis. Indeed, the main isoforms of $apoAI_M$ and $apoAI_P$ had one different charge as compared to normal apoAI in that mutant apoAI and the main isoform of normal apoAI have pI at 5.52 and 5.64, respectively. The gel slices containing the apolipoproteins of interest were vacuum dried.

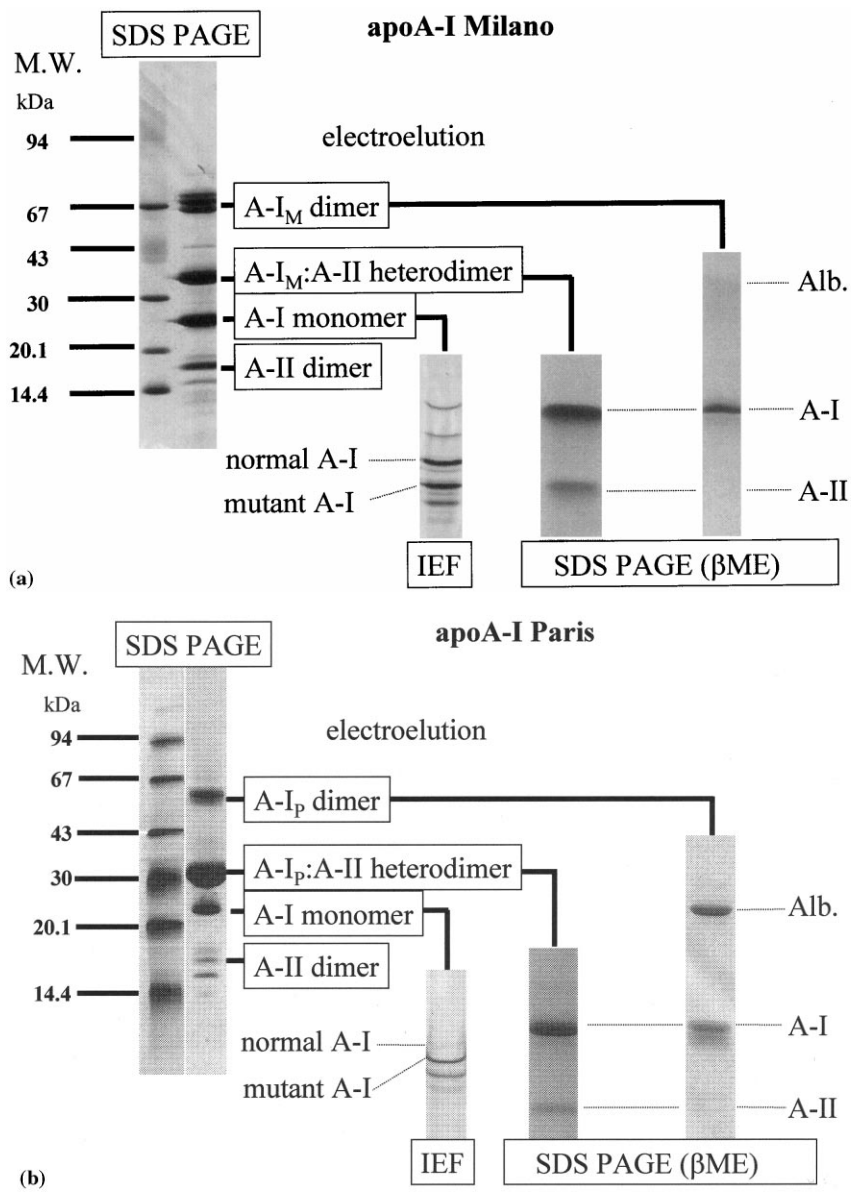


Fig. 1. Separation of mutant and normal apoAI and apoAII included in mutant apoAI dimer, mutant apoAI:AII heterodimer, apoAI monomer and apoAII dimer from HDL purified by sequential ultracentrifugation from $apoAI_M$ (a) and $apoAI_P$ (b) subjects. First, the different apoAI- and apoAII-containing complexes in $apoAI_M$ (one representative of the two subjects) (a) and the $apoAI_P$ subject (b) were separated by SDS-PAGE in non reducing conditions. The different bands were immediately detected after completion of electrophoresis and electroeluted (see Section 2). These figures show a Coomassie Blue stained gel for illustration. The electroeluted complexes were applied on an SDS-Page gel in reducing conditions (SDS-PAGE, βME) (mutant apoAI dimer and mutant apoAI:AII heterodimer) or on an isoelectric focusing gel (IEF) (AI monomer) to purify mutant and/or normal apoAI and apoAII from complexes. These gels were stained with Coomassie Blue.

2.3. Preparation for gas chromatography/mass spectrometry (GC/MS) analysis

The dried protein-containing gel slices were hydrolyzed in 6 N Ultrex HCl (JT Baker, Phillipsburg, USA) for 24 h at 110°C on a heating block. Hydrolysates were centrifuged (3000 rev./min) for 30 min. Supernatants were evaporated under nitrogen and samples were redissolved in 1 ml of 1 N acetic acid. The acid solution was applied to a 1 ml AG-50W*8 (H^+ form) cation exchange resin (Bio-Rad Laboratories, Hercules, CA, # 143-5441). After washing resin with deionized water, amino acids were eluted with 3 ml of 3 M NH_4OH. The elutes containing amino acids were collected in vials and dried in a speed-vac concentrator.

The dry amino acids were esterified by adding 0.5 ml of a fresh acetyl chloride/n-propanol (1:5, v/v) solution and then heated at 110°C for 1 h. After cooling, samples were evaporated under nitrogen and 100 μl of heptafluorobutyric anhydride (HFBA) was added to each sample. The samples were heated at 60°C for 20 min, gently dried under nitrogen, dissolved in HPLC grade ethyl acetate and sealed in an autosampler vial.

2.4. Isotopic abundance measurements of leucine

Samples were analyzed by methane negative chemical ionisation GC-MS using a Finnigan-MAT SSQ 710 B instrument (Finnigan-MAT, San José, USA). Chromatographic separation was performed using a 30 m × 25 mm DB1 capillary column. The flow rate was 1 ml/min and the temperature was programmed from 75 to 280°C at 10°C/min. Isotope ratios were determined in duplicate with a third measurement if the duplicate measurements differed by over 5%. Selected charged ions with a m/z of 349 for leucine and 352 for deuterated leucine were monitored. Calibration standards containing known amounts of D3-leucine in the range of 0–10% enrichment were analyzed by GC/MS. Standard curves were analyzed by linear regression and were used to obtain isotope ratios from observed ion current ratio data of unknowns. Isotope ratios (R) were converted to enrichment values (E) by calculating fractional abundances such as $q = R/(R+1)$ and subtracting the contribution of natural material (q_n) to yield $E_t = q_t - q_n$. The enrichment was then converted to tracer-to-tracee ratio by using the formula $e_t/(e_i - e_t)$, where e_t is the enrichment of each sample at time t and e_i is the enrichment of the infusate (0.99 for the deuterated leucine used in this study) [17].

2.5. Determination of kinetic parameters

The tracer-to-tracee ratios of VLDL-apo B100 and different purified HDL-apolipoproteins were fitted to monoexponential functions using SAAM II (SAAM Institute, University of Washington, Seattle, WA). The function was defined as: $A_t = A_p \times (1 - e^{-k(t-d)})$, where A_t is the tracer-to-tracee ratio at time t, A_p that of the tissue precursor amino acid pool from which the protein was derived, k the fractional synthetic rate (FSR), and d the delay time. For each apoAI and apoAII species, the tracer-to-tracee ratio of the precursor pool was estimated from the VLDL-apo B100 enrichment at the plateau of the tracer-to-tracee ratio [18]. When the subjects are under steady state conditions, the FSR is equal to the fractional catabolic rate (FCR). Apolipoprotein production rates (PR) were determined using the formula: PR = (FSR × plasma apolipoprotein concentration × plasma volume)/(body wt). The PR of apoAI and apoAII were calculated as the sum of the PR of the different species of the respective apolipoproteins. Plasma volume was assumed to be 0.04 l/kg body wt.

2.6. Analytical methods

Cholesterol and triglycerides were determined by enzymatic procedures (Boehringer Mannheim, Germany) adapted to a Hitachi 705 analyser. Cholesterol was measured (Boehringer Mannheim, Germany) in HDL-containing supernatant after phosphotungstate/magnesium chloride precipitation of apo B-containing lipoproteins and in VLDL isolated by ultracentrifugation [14]. LDL-cholesterol was calculated by subtracting VLDL-C and HDL-C from total cholesterol. Total apoAI and apo B were quantified by laser-immunonephelometry (Behring, Marburg, Germany). Total apoAII was measured by electroimmunoassay (Sebia, Issy-les-Moulineaux, France). Particles containing apoAI but free of apoAII were quantified by differential electroimmunoassay as previously described [19]. ApoAI content in particles containing both apoAI and apoAII was calculated by subtracting LpAI from total plasma apoAI. The apo E phenotype was determined from total plasma by isoelectric focusing [20]. For each subject, the amount of apoAI and apoAII in each band was checked by quantifying Coomassie Blue R250 extracted from stained bands with pyridine [21] first after electrophoresis of HDL in non-denaturing conditions and in the presence of SDS, second from bands obtained by electroelution and electrophoresis as above. The percentages of apoAI (respectively apoAII) in each complex were calculated assuming the sum of apoAI (respectively apoAII) included in the different complexes was 100%. The concentration of apoAI (respectively apoAII) in each complex was calculated by multiplying the percentage of apoAI (respectively apoAII) in each band by apoAI (respectively apoAII) plasma concentration.

Table 1
Plasma lipids, apolipoproteins and lipoprotein particles in patients and control subjects[a]

	ApoAI Milano		ApoAI Paris	Control subjects*
	Patient 1	Patient 2	Patient 3	
Cholesterol	203	123	195	233 ± 46
Triglycerides	251	100	167	153 ± 68
VLDL-chol.	24	16	27	22 ± 15
LDL-chol.	164	125	158	155 ± 44
HDL-chol.	15	14	10	56 ± 18
ApoAI	57	44	50	154 ± 28
ApoAII	20	11	11	37 ± 8
Apo B	155	86	140	124 ± 29
Lp AI	26	17	6	52 ± 16
Lp AI/AII	21	27	44	84 ± 15

[a] Values are mg/dl.
* Reference values (mean ± S.D.) are those obtained from a representative population in the north of France (the ECTIM study).

Table 2
Concentrations of apoAI and apoAII in apoAI Milano and apoAI Paris subjects in the different complexes in which these apolipoproteins were included

	ApoAI Milano		ApoAI Paris
	Patient 1	Patient 2	Patient 3
ApoAI (mg/dl) in			
Normal AI monomer	20.2	16.0	0.3
Mutant AI monomer	14.2	4.9	4.6
Mutant AI dimer	10.2	11.4	1.5
Mutant AI:AII heterodimer	12.2	11.7	43.7
ApoAII (mg/dl) in			
AII:AII dimer	10.5	6.4	4.9
Mutant AI:AII heterodimer	9.5	4.6	6.1

3. Results

The lipid status of the subjects during the kinetic study is reported in Table 1 compared to values recorded for male subjects recruited to obtain a representative control population and included in the ECTIM study [22]. Subjects 1 ($apoAI_M$) and 3 ($apoAI_P$) had a normal cholesterol level while subject 2 ($apoAI_M$) had a low total cholesterol. The plasma triglyceride levels of subjects 1 and 3 were in the upper normal range and normal in subject 2, while LDL-cholesterol in the three subjects was in the normal range. HDL cholesterol was dramatically low, below the fifth percentile of age- and sex-matched controls in all three subjects. All three subjects also had markedly reduced levels of both apoAI (57, 44 and 50 mg/dl) and apoAII (20, 11 and 11 mg/dl). LpAI and LpAI:AII levels were low, but the relative decrease in LpAI was greater in the AI_P subject than in the AI_M subjects. Indeed, the ratio between the concentrations of LpAI and LpAI:AII was normal (subject 2) or high (subject 1) in the apoAI Milano subjects, whereas this ratio was low in the $apoAI_P$ subject.

The electrophoretic profiles of HDL apolipoproteins during purification are shown in Fig. 1a,b. The two AI_M subjects had a similar pattern and so only one is represented in Fig. 1a. The distributions of the various bands corresponding to the various molecular complexes including mutant or normal apoAI with/without apoAII were clearly different between the $apoAI_M$ subjects on the one hand and the $apoAI_P$ one on the other hand. After electrophoresis of HDL proteins of $apoAI_M$ subjects in SDS-PAGE gel in non reducing conditions (Fig. 1a, left part), AI_M dimer, $apoAI_M$:AII complex and apoAI monomer appeared quantitatively similar. In contrast, in the $apoAI_P$ subject, the AI_P:apoAII heterodimer was clearly the most important band among the apoAI-containing complexes (Fig. 1b, left part). In the apoAI Milano subjects, mutant and normal apoAI monomers were both clearly identifiable. In the AI_P subject, the separation of the apoAI monomer by isoelectric focusing (Fig. 1b: IEF) showed that the mutant was predominant, the normal apoAI being detectable only in a faint band in the position of the main apoAI isoform. The plasma levels of apoAI and apoAII in the different complexes were measured in each subject three times during the kinetic study (0, 6 and 12 h). These concentrations were stable during the study and the averages of these concentrations are presented in Table 2 and confirm the qualitative data shown in gels. In AI_M, the mutant apoAI was evenly distributed between the AI_M dimer, the AI_M:AII complex and the monomer. The low levels of both normal and mutant apoAI, 16–20 and 28–37 mg/dl respectively, explained the low level of total plasma apoAI. The distribution of apoAI was totally different in apoAI Paris. Indeed, in this subject, normal apoAI was significantly lower, the major apoAI form in plasma being the AI_P:AII heterodimer. ApoAII was almost equally distributed between the apoAI:AII heterodimer and apoAII dimer in the three subjects with slightly more apoAII in the apoAII dimer in the $apoAI_M$ subjects and the opposite in the $apoAI_P$ subject.

3.1. Kinetic studies

Cholesterol, triglycerides, total apoAI and the plasma levels of mutant and normal apoAI were stable during the kinetic study in each subject (data not shown). The tracer-to-tracee ratios of free plasma leucine remained in the steady state throughout the infusion period in all subjects and that of VLDL-apo B100 reached a plateau after 5–6 h in both the carriers and controls (data not

shown). The tracer-to-tracee ratios of apoAI and apoAII purified from the different complexes were measured as described in Section 2 and are presented as a function of time, together with curves best fitting the exponential function (Fig. 2a,b). The metabolism of the various apoAI forms differed clearly between $apoAI_M$ subjects on the one hand and the $apoAI_P$ subject on the other hand (Fig. 2a). In the two $apoAI_M$ subjects, D3-leucine enrichment increased most rapidly in the normal apoAI monomer and most slowly in the $apoAI_M$ dimer. The increases in enrichment of $apoAI_M$ monomer and $apoAI_M$ in $apoAI_M$:AII heterodimer were intermediate between those of $apoAI_M$ and apoAI monomers. In the $apoAI_P$ subject, the increase in the enrichment of normal apoAI monomer was also the most notable, those of $apoAI_P$ dimer, $apoAI_P$ in $apoAI_P$:AII heterodimer and $apoAI_P$ monomer being lesser and near each other.

D3-leucine enrichment in apoAII dimer was faster than that of apoAII in apoAI:AII heterodimer in $apoAI_M$ and $apoAI_P$ subjects (Fig. 2b). However, the difference between the two curves of apoAII enrichment was greater in $apoAI_P$ than in $apoAI_M$ subjects, the enrichment in apoAII of apoAI:AII heterodimer being very slow in the $apoAI_P$ subject.

The FCR of each apolipoprotein was calculated as described in Section 2 and presented in Table 3. The data for the two $apoAI_M$ subjects were similar and differed from $apoAI_P$. A striking result was the dramatic increase in normal apoAI FCR in the three carriers as compared to that of normolipidemic subjects. Simultaneously, the FCRs of the $apoAI_M$ monomer and $apoAI_M$ in $apoAI_M$:AII heterodimer were lower than that of the normal apoAI monomer in each of the two $apoAI_M$ subjects, the former corresponding to 61–71 and the latter to 53–60% of the normal apoAI monomer FCR measured within the same subject. The FCR of $apoAI_M$ dimer was also less

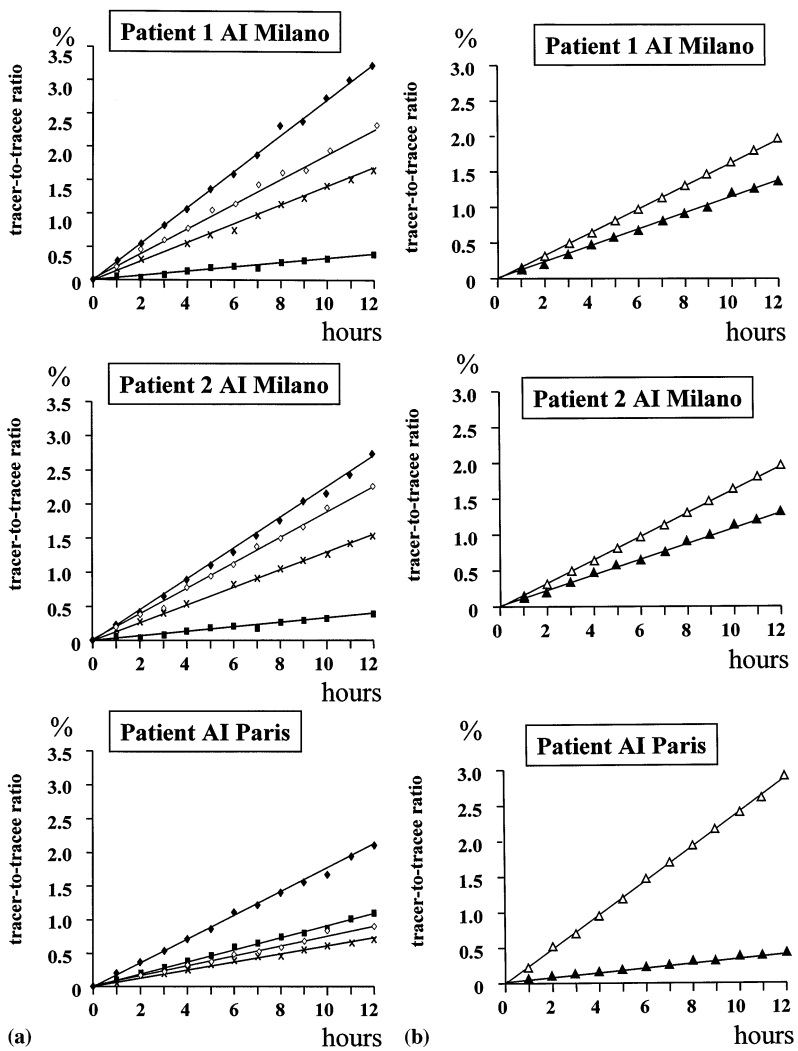


Fig. 2. Tracer-to-tracee ratio in apoAI present in homodimer (■), apoAI:AII heterodimer (×), mutant monomer (◇) and normal monomer (◆) (a) and in apoAII in apoAI:AII heterodimer (▲) and in apoAII dimer (△) (b) in apoAI Milano subjects (1 and 2) and in apoAI Paris subject.

Table 3
Fractional Catabolic Rate (FCR) and Production Rate (PR) of apoAI and apoAII in apoAI Milano and apoAI Paris subjects in the different complexes in which these apolipoproteins were included

	ApoAI Milano				ApoAI Paris		Normolipidemic subjects	
	Patient 1		Patient 2		Patient 3			
	FCR (day^{-1})	PR (mg/kg per day)	FCR (day^{-1})	PR (mg/kg per day)	FCR (day^{-1})	PR (mg/kg per day)	FCR (day^{-1})	PR (mg/kg per day)
ApoAI								
Normal AI monomer	0.805	6.50	0.671	4.29	0.526	0.05	–	–
Mutant AI monomer	0.570	3.24	0.406	0.80	0.228	0.42	–	–
Mutant AI dimer	0.114	0.47	0.118	0.54	0.263	0.16	–	–
Mutant AI:AII heterodimer	0.430	2.10	0.401	1.88	0.182	3.18	–	–
Total	–	12.3	–	7.51	–	3.81	0.190 ± 0.0015	9.01 ± 1.03
ApoAII								
AII/AII dimer	0.479	2.01	0.487	1.25	0.719	1.41	–	–
AI/AII heterodimer	0.358	1.36	0.344	0.63	0.108	0.26	–	–
Total	–	3.37	–	1.88	–	1.67	0.164 ± 0.010	1.66 ± 0.57

than that of normal apoAI monomer but at its highest point, corresponded to 14–18% of the normal apoAI monomer FCR. In the $apoAI_P$ subject, the mutant monomer FCR corresponded to 43% of the FCR of the normal monomer form. The other two forms of mutant $apoAI_P$ also had a lower FCR than the normal apoAI monomer, corresponding to 35 and 50%, for the $apoAI_P$:AII and $apoAI_P$ dimer, respectively.

In the $apoAI_M$ subjects, the apoAII FCR was higher than that measured in normolipidemic subjects, as regards both forms—homo- and heterodimer—in which apoAII was included. However, apoAII in homodimer was catabolized more rapidly than apoAII in heterodimer. In the $apoAI_P$ subject, the FCR of apoAII heterodimer was very high (about seven times that of apoAII in $apoAI_P$:AII heterodimer).

The production rate of each protein was calculated from the FCR and plasma concentration in each subject. The production rate of total apoAI (respectively apoAII) was calculated as the sum of the production rates of each form of apoAI (respectively apoAII). The production rate of apoAI in $apoAI_M$ subjects was in the normal range, whereas the production rate of apoAI was remarkably lower in the $apoAI_P$ subject than in control subjects. The production rates of apoAII were moderately elevated in subject 1 ($apoAI_M$), but normal in subject 2 ($apoAI_M$) and subject 3 ($apoAI_P$).

4. Discussion

The distribution and kinetics of the various molecular forms of apoAI and apoAII were clearly different in $apoAI_M$ and $apoAI_P$ carriers. Mutant apoAI was similarly distributed among complexes (dimers and monomer) in $apoAI_M$ subjects whereas mutant apoAI was essentially in the apoAI:AII heterodimer in the $apoAI_P$ subject. Furthermore, the concentration of normal apoAI was very low in the $apoAI_P$ subject, whereas it represented about one third of total apoAI in the $apoAI_M$ subjects. Differences were also evident for apoAII, which was found mainly as apoAII dimer in the $apoAI_M$ subjects, and as AI_P:AII complex in the $apoAI_P$ subject. These differences in distribution correlate well with the kinetic parameters, FCR and PR. The apoAI of the apoAI:AII complex and the mutant monomer were catabolized about twice as fast in $apoAI_M$ as in $apoAI_P$. On the other hand, the apoAI dimer was catabolized twice as slowly in $apoAI_M$ as in $apoAI_P$. Normal apoAI monomer was rapidly catabolized both in $apoAI_M$ and in $apoAI_P$.

A previous study carried out in two $apoAI_M$ subjects using radiolabeled apoAI [11] showed similar FCRs for normal and mutant apoAI, a result which contrasts with the findings of this study. The reasons for this discordance are not understood. This could be due to a difference in subjects included or in methodology. Two groups of $apoAI_M$ subjects were differentiated by the presence or absence of hypertriglyceridemia and the degree of hypoHDLemia [13]. Each of the two $apoAI_M$ subjects included in our study belongs to a different group on the basis of triglyceride levels, which could explain the slight difference in the normal apoAI FCR between them, but the large difference in normal apoAI FCR measured in the subjects included in the previous and present studies is not explained by this heterogeneity. In the two studies, the subjects were on ad libitum diet, but it is also unlikely that a possible variability in diet should explain such a large metabolic difference.

In the study using radiolabeled apoAI, FCR results were obtained without separating apoAI-containing lipoproteins. So, the FCR of $apoAI_M$ was measured as if mutant apoAI was in a metabolically homogenous pool. We demonstrated in our study that this is not true as the metabolism of $apoAI_M$ present in the different complexes is quite different. However, we have been able to calculate $apoAI_M$ FCR from our data (by adding the PR of the $apoAI_M$ contained in the different complexes and dividing the total PR by the $apoAI_M$ plasma level) as if $apoAI_M$ had been purified from total HDL (as in Ref. [11]), i.e. without distinguishing the different complexes. The $apoAI_M$ FCR were 0.397 and 0.288 day^{-1} in the two $apoAI_M$ carriers, values similar to those previously reported (0.319 and 0.380 day^{-1}).

Mutant apoAI contained in the different complexes clearly had a different metabolism. This finding was also suggested by a previous kinetic study carried out in $apoAI_M$ subjects which showed that $apoAI_M$ was catabolized more rapidly as a monomer than as a dimer [11], but this study did not differentiate hetero- and homodimers. Both the present kinetic study and the previous one on $apoAI_M$ subjects analyzed data on the assumption that there is little or no exchange of these apolipoproteins between complexes during the time of study. Though there is no direct demonstration for that, the obvious differences in the FCR of mutant apoAI present in the different complexes suggest that this hypothesis is confirmed. Indeed, if fast exchange was present between the complexes containing apoAI, the enrichment in D3-leucine would have been similar in mutant apoAI purified from the different complexes. So we can calculate the synthesis of each protein as the sum of the production rates of apoAI in the different complexes. Nevertheless, in spite of very different normal apoAI FCR and differences in the measurement of $apoAI_M$ FCR in the two studies using radio- and stable isotope labelling of apoAI, the total production rates of apoAI were similar between them and normal compared to apoAI synthesis in normolipidemic subjects.

So, our results go beyond those of the kinetic study using radiolabeled apoAI obtained in $apoAI_M$ subjects by showing a clear difference in the metabolism be-

tween hetero- ($apoAI_M$:AII) and homodimers ($apoAI_M$:AI_M), the latter being more slowly catabolized than the former. The two types of dimers are present on the surface of different types of particles, the $apoAI_M$:AII complex being in LpAI:AII and homodimers essentially in LpAI [23].

Furthermore, Franceschini et al. demonstrated that the presence of disulfide-linked dimers increases HDL stability creating a more resistant state for the remodeling by LCAT and CETP [24]. Thus, the plasticity of HDL which is a key factor in their metabolism is restricted. The increased stability of $apoAI_M$ HDL appears essentially due to homodimers. Indeed, homodimers remarkably limit HDL heterogeneity [25] and HDL containing homodimers are less reactive to LCAT [26]. On the other hand, the low $apoAI_M$ dimer FCR, suggesting an increase in the in vivo stability of HDL containing homodimers, is coherent with this in vitro finding. Conversely, although heterodimers have not been analyzed yet, the larger heterodimer FCR suggests that this dimer does not have the same properties as the heterodimer.

Although $apoAI_P$ forms disulfide-linked dimers such as $apoAI_M$, the difference in FCR between $apoAI_M$ and $apoAI_P$ suggests that the biochemical properties of $apoAI_P$ and $apoAI_M$ dimers are different. The low plasma level of apoAI in the $apoAI_P$ subject was mainly due to an abnormal metabolism of normal apoAI which has a high FCR and a very low synthesis.

The very low production rate of normal apoAI in $apoAI_P$ might be explained by an additional abnormality of the normal apoAI gene. Although the sequence of the four exons of this gene are apparently normal (personal communication), we cannot confirm this hypothesis because the gene, particularly its promoter region, has not been fully investigated yet. Other genes such as apoAI could also lower apoAI synthesis. Unfortunately, we have not been able to carry out a kinetic study on other members of the family, particularly those with a moderately low HDL level, who all refused the procedure. However, analysis of the family tree is suggestive of this hypothesis since the parents' proband has a low HDL level for both the father who is carrying the $apoAI_P$ gene and the mother who is not [9]. So, the patient could have two abnormalities, $apoAI_P$ and another as yet unidentified defect, inherited from his father and his mother respectively, both lowering plasma apoAI level. Neither can we exclude the possibility that the low synthesis of apoAI is only caused by environmental factors, but the very low level of normal apoAI synthesis argues against this hypothesis.

The low apoAII levels in $apoAI_M$ subjects were due to an accelerated catabolism of both heterodimers (apoAI:AII) and apoAII dimers. However, the FCR of these dimers was lower than those of the apoAII monomer and the $apoAI_M$:AII heterodimer, which explains the smaller decrease in apoAII than in apoAI plasma levels. Conversely, the low apoAII level in the $apoAI_P$ subject was due to a high FCR of apoAII homodimer while apoAII FCR in heterodimer was very low. As no physicochemical study was carried out to analyze the lipid binding of the $apoAI_P$-containing complexes, no explanation for the difference between FCR levels can be given. It only suggests a modification in lipid interaction with apoAII when apoAII is included in the heterodimer compared to the homodimer.

In conclusion, features common to both $apoAI_M$ and $apoAI_P$ are the production of low HDL levels and the possibility of establishing dimers in relation to the presence of a cysteine in the mutant protein. However, the structural alteration is not similar in the two types of mutation as shown by the different metabolism of the apoAI- and apoAII-containing complexes which is due to the properties of these complexes.

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